Metro Transportation Analytics

1. Project Overview

This project analyzes Metro transaction data to understand passenger activity, fare revenue, station and line performance, travel-time patterns, and passenger/payment segments. The workflow connects **E xcel → Python → S QL → Power B I** to turn transaction-level data into business insights.

2. Dataset Summary

- **Records:** 19,998
- **Distinct transaction IDs:** 19,997
- **Period:** 1 January–31 December 2025
- **Entry/exit stations:** 46 each
- **Metro lines:** R ed and Blue
- **Average fare:** 36.05
- **Total fare value:** 720,903.76
- **Average journey duration:** 36.19 minutes

3. Data Preparation and Python Analysis

Excel was used for initial review and validation. Python/pandas was used to profile the dataset, prepare derived fields, and summarize demand and fare patterns.

The analytical C S V includes journey details, fare and distance, passenger and ticket attributes, payment mode, date/time features, and data-quality flags.

Data-quality items to review:

- One duplicate transaction ID.
- Two non-positive fare values.

- Two ages outside a basic 0–120 range.
- Inconsistent peak-hour fields: Peak flag, `Peak_Category`, and `PeakFlag` do not agree.
- The Revenue column has a leading space and should be checked against `Fare`.

These issues should be investigated and resolved or explicitly documented before final reporting.

4. SQL Business Analysis

The SQL script is intended to analyze:

- Total and distinct transactions, fare totals, and average fare.
- Daily and hourly demand.
- Top entry stations and Metro line performance.
- Peak versus off-peak activity.
- Passenger, ticket, and payment-mode segments.
- Station activity and route patterns.

5. Key Findings

Based on `Metro_cmpt _cleaned_data_fixed. c s v`:

- **Line performance:** R ed recorded 10,806 transactions and fare value of 389,665.54; Blue recorded 9,192 transactions and 331,238.22.
- **Top entry station:** HITEC City had 627 records, followed by KPHB Colony (607) and MG Bus station (604).
- **Payment mode:** UPI was most common with 8,293 records, followed by S mart Card (5,967).
- **Passenger type:** Adult was the largest group with 13,585 records.
- **Ticket type:** Single Journey was the most common ticket type with 11,014 records.
- **Hourly activity:** hour 10 had the most records (1,815), followed by hour 21 (1,736) and hour 20 (1,733).

6. Power BI Dashboard

The project R E ADME and guide describe a three-page management report:

1. **Executive Overview:** total trips, fare value, average fare/duration, monthly trend, line revenue, top stations, and filters.
2. **Station & Route Performance:** station rankings, line comparison, station-by-line matrix, and trips-versus-revenue analysis.
3. **Operations & Customer Analysis:** hourly demand, peak periods, weekday/weekend activity, passenger types, ticket types, and payment modes.

7. Business Recommendations

- Resolve duplicate IDs, invalid fares, implausible ages, and duration validation errors.
- Create one agreed peak/off-peak rule and apply it consistently across Python, SQL, and Power BI.
- Investigate high-volume stations using entry and exit counts plus actual capacity and service data before proposing operational changes.
- Compare line-level transaction counts, average fare, distance, and duration before interpreting fare differences.
- Monitor payment-mode mix and use additional settlement or failed-payment data if available.
- Reconcile row counts, distinct IDs, date range, and fare totals a teach stage of the pipeline.

Conclusion

This project demonstrates an end-to-end analytics workflow using Excel, Python, SQL, and Power BI. The supplied dataset supports descriptive analysis of transaction volume, fare value, station activity, line performance, hourly demand, and customer/payment segments.